

Example

Evaluate the integral

$$\oint_C \frac{1}{z - z_0} dz,$$

where C is a circle centered at z_0 and of any radius. The path is traced out once in the anticlockwise direction.

Solution

The circle can be parameterized by

$$z(t) = z_0 + re^{it}, \quad 0 \leq t \leq 2\pi,$$

where r is any positive real number. The contour integral becomes

$$\oint_C \frac{1}{z - z_0} dz = \int_0^{2\pi} \frac{1}{z(t) - z_0} \frac{dz(t)}{dt} dt = \int_0^{2\pi} \frac{ire^{it}}{re^{it}} dt = 2\pi i.$$

The value of the integral is independent of the radius r .

Example

Evaluate the integral

$$(i) \int_C |z|^2 dz \text{ and } (ii) \int_C \frac{1}{z^2} dz,$$

where the contour C is

- (a) the line segment with initial point -1 and final point i ;
- (b) the arc of the unit circle $\operatorname{Im} z \geq 0$ with initial point -1 and final point i .

Do the two results agree?

Solution

(i) Consider $\int_C |z|^2 dz$,

(a) Parameterize the line segment by

$$z = -1 + (1 + i)t, \quad 0 \leq t \leq 1,$$

so that

$$|z|^2 = (-1 + t)^2 + t^2 \quad \text{and} \quad dz = (1 + i) dt.$$

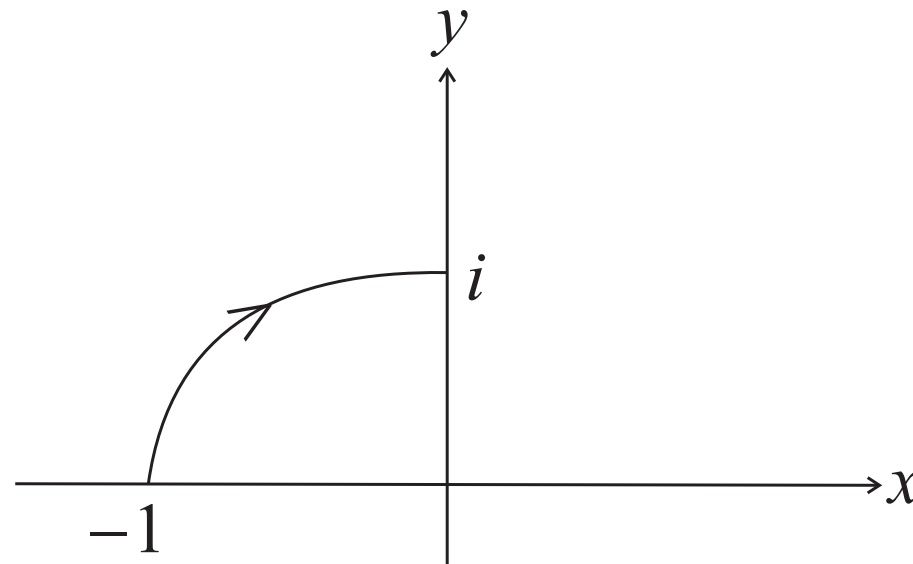
The value of the integral becomes

$$\int_C |z|^2 dz = \int_0^1 (2t^2 - 2t + 1)(1 + i) dt = \frac{2}{3}(1 + i).$$

(b) Along the unit circle, $|z| = 1$ and $z = e^{i\theta}$, $dz = ie^{i\theta}d\theta$. The initial point and the final point of the path correspond to $\theta = \pi$ and $\theta = \frac{\pi}{2}$, respectively. The contour integral can be evaluated as

$$\int_C |z|^2 dz = \int_{\pi}^{\frac{\pi}{2}} ie^{i\theta} d\theta = e^{i\theta} \Big|_{\pi}^{\frac{\pi}{2}} = 1 + i.$$

The results in (a) and (b) do not agree. Hence, the value of this contour integral does depend on the path of integration.



(ii) Consider $\int_C \frac{1}{z^2} dz$.

(a) line segment from -1 to i

$$\int_C \frac{1}{z^2} dz = \int_0^1 \frac{1+i}{[-1+(1+i)t]^2} dt = -\frac{1}{-1+(1+i)t} \Big|_0^1 = -1 - \frac{1}{i} = -1 + i.$$

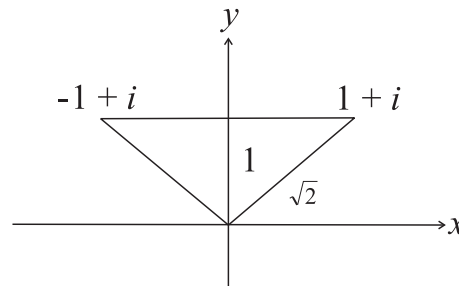
(b) subarc from -1 to i

$$\int_C \frac{1}{z^2} dz = \int_{\pi}^{\frac{\pi}{2}} \frac{1}{e^{2i\theta}} i e^{i\theta} d\theta = -e^{-i\theta} \Big|_{\pi}^{\frac{\pi}{2}} = -1 + i.$$

Example

Show that

$$\left| \int_C \frac{1}{z^2} dz \right| \leq 2, \text{ where } C \text{ is the line segment joining } -1 + i \text{ and } 1 + i.$$



Solution

Along the contour C , we have $z = x + i$, $-1 \leq x \leq 1$, so that $1 \leq |z| \leq \sqrt{2}$. Correspondingly, $\frac{1}{2} \leq \frac{1}{|z|^2} \leq 1$. Here, $M = \max_{z \in C} \frac{1}{|z|^2} = 1$ and the arc length $L = 2$. We have

$$\left| \int_C \frac{1}{z^2} dz \right| \leq ML = 2.$$

Example

Estimate an upper bound of the modulus of the integral

$$I = \int_C \frac{\text{Log } z}{z - 4i} dz$$

where C is the circle $|z| = 3$.

$$\text{Now, } \left| \frac{\text{Log } z}{z - 4i} \right| \leq \frac{|\ln |z|| + |\text{Arg } z|}{||z| - |4i||} \text{ so that}$$

$$\max_{z \in C} \left| \frac{\text{Log } z}{z - 4i} \right| \leq \frac{\ln 3 + \pi}{|3 - 4|} = \ln 3 + \pi; \quad L = (2\pi)(3) = 6\pi.$$

$$\text{Hence, } \left| \int_C \frac{\text{Log } z}{z - 4i} dz \right| \leq 6\pi(\pi + \ln 3).$$

Example

Find an upper bound for $\left| \int_{\Gamma} e^z / (z^2 + 1) dz \right|$, where Γ is the circle $|z| = 2$ traversed once in the counterclockwise direction.

Solution

The path of integration has length $L = 4\pi$. Next we seek an upper bound M for the function $e^z / (z^2 + 1)$ when $|z| = 2$. Writing $z = x + iy$, we have

$$|e^z| = |e^{x+iy}| = e^x \leq e^2, \quad \text{for } |z| = \sqrt{x^2 + y^2} = 2,$$

and by the triangle inequality

$$|z^2 + 1| \geq |z|^2 - 1 = 4 - 1 = 3 \quad \text{for } |z| = 2.$$

Hence, $|e^z / (z^2 + 1)| \leq e^2/3$ for $|z| = 2$, and so

$$\left| \int_{\Gamma} \frac{e^z}{z^2 + 1} dz \right| \leq \frac{e^2}{3} \cdot 4\pi.$$

Example

If C is the curve $y = x^3 - 3x^2 + 4x - 1$ joining points $(1, 1)$ and $(2, 3)$, find the value of

$$\int_C (12z^2 - 4iz) \, dz.$$

Method 1. The integral is independent of the path joining $(1, 1)$ and $(2, 3)$. Hence any path can be chosen. In particular, let us choose the straight line paths from $(1, 1)$ to $(2, 1)$ and then from $(2, 1)$ to $(2, 3)$.

Case 1 Along the path from $(1, 1)$ to $(2, 1)$, $y = 1$, $dy = 0$ so that $z = x + iy = x + i$, $dz = dx$. Then the integral equals

$$\int_1^2 \{12(x + i)^2 - 4i(x + i)\} \, dx = \{4(x + i)^3 - 2i(x + i)^2\} \Big|_1^2 = 20 + 30i.$$

Case 2 Along the path from $(2, 1)$ to $(2, 3)$, $x = 2$, $dx = 0$ so that $z = x + iy = 2 + iy$, $dz = i dy$. Then the integral equals

$$\int_1^3 \{12(2+iy)^2 - 4i(2+iy)\} i dy = \{4(2+iy)^3 - 2i(2+iy)^2\} \Big|_1^3 = -176 + 8i.$$

Then adding, the required value $= (20 + 30i) + (-176 + 8i) = -156 + 38i$.

Method 2. The given integral equals

$$\int_{1+i}^{2+3i} (12z^2 - 4iz) dz = (4z^3 - 2iz^2) \Big|_{1+i}^{2+3i} = -156 + 38i.$$

It is clear that Method 2 is easier.